

# JOHN ANDREI MERONG MARTINEZ

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## SUMMARY

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AI/ML engineering student who ships end-to-end systems: agentic AI, retrieval-augmented generation (RAG), LLM fine-tuning, forecasting and computer vision, taken from dataset to deployed service. Twenty personal projects, eighteen with live demos. Dean's Lister 2023-2026.

## SKILLS

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**Languages:** Python, SQL, JavaScript, Java, C, C++

**AI/ML:** PyTorch, scikit-learn, LLM Fine-Tuning (LoRA, QLoRA, Unsloth), RAG, LangChain, Agentic AI, Ollama, Transformers, CNNs, Computer Vision, Time-Series Forecasting, Model Evaluation

**Data:** pandas, NumPy, DuckDB, PostgreSQL, MongoDB, SQLite, MySQL, Tableau, Power BI

**Backend & Tools:** FastAPI, Node.js, React, Docker, Git/GitHub, GitHub Actions, ONNX, n8n, Kaggle

**Cloud:** Azure OpenAI, Google Cloud, Oracle Cloud, AWS, Vercel, Supabase, Firebase

## PROJECTS

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### YODA - Your Offline Data Agent

2026

*Tech: Python, Ollama, FastAPI, pandas, Agentic AI*

- Built a local-first data agent that answers questions over CSV and Excel files with no data leaving the machine, running a local Ollama model behind a FastAPI service.
- Designed a tool-calling loop that plans pandas operations, executes them, and validates its own output before answering.

### Aegix AI - Contract Compliance Screening

2026

*Tech: Python, LangChain, Azure OpenAI, Vector Search, RAG, FastAPI*

- Shipped a retrieval-augmented screening tool that flags non-compliant contract clauses and cites the source paragraph behind every flag.
- Built the vector-search retrieval layer over Azure OpenAI embeddings and exposed it as a FastAPI service with a React front end.

### Refusal Calibration - LLM Fine-Tuning

2025

*Tech: PyTorch, LoRA, QLoRA, Model Evaluation, Python*

- Fine-tuned a base model with LoRA so it refuses out-of-scope questions instead of answering them confidently.
- Built a held-out evaluation set that measures over-refusal as well as refusal, so the model is not rewarded for simply declining everything.

### APAW - Dam Level and Spill-Risk Nowcaster

2025

*Tech: Python, River, Online Learning, Time-Series, GitHub Actions*

- Deployed a self-updating forecaster that retrains on new Open-Meteo readings through a scheduled GitHub Actions job, running unattended.
- Used online learning so the model adapts to seasonal shift without a full retrain from scratch.

### FORGE - Fake Or Real Image Examiner

2025

*Tech: PyTorch, ONNX, Computer Vision, GitHub Actions, JavaScript*

- Trained an image classifier that separates AI-generated images from photographs, exported to ONNX so it runs in the browser with no server.
- Automated retraining and redeployment through GitHub Actions.

### Which Page Do You Fix First? - Refresh Priority on Search Data

2026

*Tech: scikit-learn, DuckDB, pandas, Causal Inference, Python*

- Ranked which pages to refresh first from real search-console data, reaching 0.88 Precision@50 against a 0.86 recency baseline.
- Ran a leakage audit and bootstrap confidence intervals so the gain was shown to be real rather than an artefact of the split.

## EXPERIENCE

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### ML Engineering Intern - FlyRank AI

2026 - Present

- Prepare and validate datasets, support model testing, and automate workflow steps in Python.
- Build data pipelines and internal automation tools that cut manual handling out of the team's day-to-day work.

### Virtual Assistant - Freelance

2025 - 2026

- Ran end-to-end recruitment, data administration and digital record keeping, improving hiring turnaround and record accuracy.

EDUCATION

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**BS Computer Science, specializing in AI/ML Engineering - Our Lady of Fatima University** 2023 - Present

- Dean's Lister 2023-2026, with placements in Python, Database Design and C Programming.

CERTIFICATIONS

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|---------------------------------------------------------------------------------------------------|-----------|
| <b>Google Advanced Data Analytics Professional Certificate</b><br><i>Coursera</i>                 | July 2026 |
| <b>Google AI Professional Certificate</b><br><i>Coursera</i>                                      | June 2026 |
| <b>AWS AI Practitioner</b><br><i>Udemy</i>                                                        | June 2026 |
| <b>Building with the Claude API</b><br><i>Anthropic</i>                                           | 2026      |
| <b>Introduction to Model Context Protocol</b><br><i>Anthropic</i>                                 | 2026      |
| <b>Oracle Cloud Infrastructure Certified AI Foundations Associate</b><br><i>Oracle University</i> | July 2025 |
| <b>Google Data Analytics Professional Certificate</b><br><i>Coursera</i>                          | May 2025  |

EXTRACURRICULAR ACTIVITIES

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Technical Team Member, English Society. Directorate Member, Junior Philippine Computer Society (JPCS).